Notes on Multiple Linear Regression

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I. Building a model

- Define the problem and collect data.
- Preprocess the data (handle missing values, encode categorical variables, etc.).
- Split the data into training and testing sets.
- Train the linear regression model on the training data.
- Evaluate the model using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), or R-squared on the testing data.
- Interpret the coefficients to understand the relationship between independent and dependent variables.

Backward Elimination

- Step 1: Select a significance level (e.g., 0.05) to stay in the model.
- Step 2: Fit the model and calculate the p-values for each coefficient.
- Step 3: Remove the variable with the highest p-value (greater than a significance level, e.g., 0.05).
- Step 4: Repeat the process until all remaining variables have p-values below the significance level.
- Step 5: The final model will contain only the statistically significant variables. Fit the model without the variables that were removed and evaluate its performance.

Forward Selection

- Step 1: Select a significance level (e.g., 0.05) to enter the model.
- Step 2: Fit the model with each independent variable separately and calculate the p-values.
- Step 3: Add the variable with the lowest p-value (below the significance level) to the model.
- Step 4: Repeat the process by adding variables one at a time, checking the p-values at each step.
- Step 5: The final model will contain only the statistically significant variables. Fit the model with the selected variables and evaluate its performance.

Bidirectional Elimination

- Step 1: Select a significance level (e.g., 0.05) for both entering and staying in the model.
- Step 2: Start with an empty model and perform forward selection to add variables based on their p-values.
- Step 3: After adding a variable, check the p-values of all variables in the model. If any variable has a p-value greater than the significance level, remove it from the model.
- Step 4: Repeat the process of adding and removing variables until no more variables can be added or removed based on the significance level.
- Step 5: The final model will contain only the statistically significant variables. Fit the model with the selected variables and evaluate its performance.

All possible models

- Step 1: Generate all possible combinations of independent variables.
- Step 2: Fit a linear regression model for each combination of variables and calculate the performance metrics (e.g., R-squared, Adjusted R-squared, AIC, BIC).
- Step 3: Compare the performance metrics across all models to identify the best-performing model.
- Step 4: Select the model with the best performance metric (e.g., highest Adjusted R-squared or lowest AIC/BIC) as the final model.
- Step 5: Fit the final model with the selected variables and evaluate its performance on the testing data.